

The Brunn-Minkowski Inequality

following R. J. Gardner

The most elementary Brunn-Minkowski inequality is that on $\mathbb{R}$.

THM 1. BRUNN-MINKOWSKI INEQUALITY ON $\mathbb{R}$.

Let $\lambda \in (0, 1)$ and $X, Y \subseteq \mathbb{R}$ be nonempty bounded measurable sets s.t. $(1 - \lambda)X + \lambda Y$ is also measurable. Then,

$$\text{vol}((1 - \lambda)X + \lambda Y) \geq (1 - \lambda) \text{vol } X + \lambda \text{vol } Y$$

Proof.

W.l.o.g. let X, Y be compact and the general noncompact case follows by using a compact exhaustion. We first prove that $X + Y$ is compact. Let $\{x_n + y_n\}_{n \in \mathbb{N}}$ be a sequence of $X + Y$ and we consider $\{x_n\}_{n \in \mathbb{N}} \subseteq X$ and $\{y_n\}_{n \in \mathbb{N}} \subseteq Y$. By compactness of X there exists a convergent subsequence $\{x_n\}_{n \in A} \subseteq \{x_n\}_{n \in \mathbb{N}} \rightarrow x \in X$, and then by compactness of Y there exists a convergent subsequence $\{y_n\}_{n \in B} \subseteq \{y_n\}_{n \in A} \rightarrow y \in Y$. Hence $\{x_n + y_n\}_{n \in B} \rightarrow x + y \in X + Y$ is a convergent subsequence. As measures of sets are translation-invariant we may w.l.o.g. translate $X \subseteq \mathbb{R}^-$ and $Y \subseteq \mathbb{R}^+$ with $X \cap Y = \{0\}$. Using the origin we have $X + Y \supseteq X \cup Y$. Dilations preserve this relation by scaling elements on both sides $(1 - \lambda)X + \lambda Y \supseteq (1 - \lambda)X \cup \lambda Y$. Hence,

$$\text{vol}((1 - \lambda)X + \lambda Y) \geq \text{vol}((1 - \lambda)X \cup \lambda Y) = (1 - \lambda) \text{vol } X + \lambda \text{vol } Y$$

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There are several variants and generalizations of this classic theorem.

We preliminarily define several objects that we will use extensively.

DEF 1. Let $\mathbb{B}^n$ denote the unit n -ball of $\mathbb{R}^n$ and $\mathbb{S}^{n-1} = \partial \mathbb{B}^n$ the unit $(n - 1)$ -sphere. For $\mathbf{u} \in \mathbb{S}^{n-1}$ denote $\mathbf{u}^\perp$ to be the hyperplane tangent to $\mathbb{S}^{n-1}$ and normal to $\mathbf{u}$. We identify the k -dimensional Lebesgue measure vol_k with the k -dimensional Hausdorff measure, and so integration w.r.t. $d\mathbf{x}$ will denote integration w.r.t vol_k (for the appropriate k) and $d\mathbf{u}$ will denote integration w.r.t. vol_{n-1} .

DEF 2. For $X, Y \subseteq \mathbb{R}^n$ their Minkowski sum is defined to be,

$$X + Y = \{\mathbf{x} + \mathbf{y} \in \mathbb{R}^n \mid \mathbf{x} \in X, \mathbf{y} \in Y\}$$

Similarly for $\lambda \in \mathbb{R}$ their ‘dilation’ (if positive) or ‘reflection then dilation’ (if negative) is defined,

$$\lambda X = \{\lambda \mathbf{x} \mid \mathbf{x} \in X\}$$

When $\lambda \in (0, 1)$ the set $(1 - \lambda)X + \lambda Y$ is called the ‘convex combination of X and Y ’.

DEF 3. Define the surface area of a set $X \subseteq \mathbb{R}^n$ to be,

$$\text{sur } X = \lim_{\varepsilon \rightarrow 0} \frac{\text{vol}(X + \varepsilon \mathbb{B}^n) - \text{vol } X}{\varepsilon}$$

We use this definition when X is a convex body or is compact with piecewise C^1 boundary.

DEF 4. A function f is concave on a convex set X if,

$$f((1 - \lambda)\mathbf{x} + \lambda\mathbf{y}) \geq (1 - \lambda)f(\mathbf{x}) + \lambda f(\mathbf{y})$$

For all $\mathbf{x}, \mathbf{y} \in X$. This is equivalent to that $-f$ is convex. A nonnegative function is ‘log-concave’ if its logarithm is concave, or by exponentiating,

$$f((1 - \lambda)\mathbf{x} + \lambda\mathbf{y}) \geq f(\mathbf{x})^{1-\lambda} f(\mathbf{y})^\lambda$$

And by the AM-GM inequality if a function is concave then it is log-concave.

DEF 5. Define the level-set of a nonnegative measurable function and a value,

$$\{f \geq t\} = \{\mathbf{x} \in \mathbb{R}^n \mid f(\mathbf{x}) \geq t\}$$

We have by Fubini’s Theorem,

$$\int_{\mathbb{R}^n} f(\mathbf{x}) \, d\mathbf{x} = \int_{\mathbb{R}^n} \int_0^{f(\mathbf{x})} dt \, d\mathbf{x} = \int_0^\infty \int_{\{f \geq t\}} d\mathbf{x} \, dt = \int_0^\infty \text{vol}\{f \geq t\} \, dt$$

The intermediate equality,

$$f(\mathbf{x}) = \int_0^\infty 1_{\{f \geq t\}}(\mathbf{x}) \, dt$$

Is known as the layer-cake representation of f .

The next result is an analytic inequality instrumental to proving Brunn-Minkowski,

THM 2. PREKOPA-LEINDLER INEQUALITY ON $\mathbb{R}$

Let $\lambda \in (0, 1)$ and f, g, h be nonnegative integrable functions on $\mathbb{R}$ satisfying,

$$h((1 - \lambda)x + \lambda y) \geq f(x)^{1-\lambda} g(y)^\lambda$$

For all $x, y \in \mathbb{R}$. Then,

$$\int_{\mathbb{R}} h \geq \left(\int_{\mathbb{R}} f \right)^{1-\lambda} \left(\int_{\mathbb{R}} g \right)^\lambda$$

Proof.

W.l.o.g. we normalize f and g by their supremum so that,

$$\sup_{\mathbb{R}} f = \sup_{\mathbb{R}} g = 1$$

Note for $t \in \mathbb{R}^+$ that $f(x), g(y) \geq t$ jointly means $h((1 - \lambda)x + \lambda y) \geq t$ by the assumption for all λ . Written in terms of level sets,

$$\{h \geq t\} \supseteq (1 - \lambda)\{f \geq t\} + \lambda\{g \geq t\}$$

Therefore,

$$\begin{aligned} \int_{\mathbb{R}} h &\geq \int_0^1 \text{vol}\{h \geq t\} dt \\ &\geq \int_0^1 \text{vol}((1 - \lambda)\{f \geq t\} + \lambda\{g \geq t\}) dt \\ &\geq \int_0^1 (1 - \lambda) \text{vol}\{f \geq t\} + \lambda \text{vol}\{g \geq t\} dt \\ &= (1 - \lambda) \int_0^1 f + \lambda \int_0^1 g \\ &= \left(\int_0^1 f \right)^{1-\lambda} \left(\int_0^1 g \right)^{\lambda} \end{aligned}$$

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The n -dimensional case uses the one-dimensional case in induction.

THM 3. PREKOPA-LEINDLER INEQUALITY ON $\mathbb{R}^n$

Let f, g, h be nonnegative measurable functions on $\mathbb{R}^n$ satisfying,

$$h((1 - \lambda)\mathbf{x} + \lambda\mathbf{y}) \geq f(\mathbf{x})^{1-\lambda} g(\mathbf{y})^{\lambda}$$

For all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. Then,

$$\int_{\mathbb{R}^n} h \geq \left(\int_{\mathbb{R}^n} f \right)^{1-\lambda} \left(\int_{\mathbb{R}^n} g \right)^{\lambda}$$

Proof.

The proof proceeds by induction on n . The base case is given by THM 2. Suppose true for all less than n . For all $s \in \mathbb{R}$ define respectively,

$$f_s, g_s, h_s: \mathbb{R}^{n-1} \rightarrow \mathbb{R}, \quad \mathbf{z} \mapsto f(\mathbf{z}, s), g(\mathbf{z}, s), h(\mathbf{z}, s)$$

For $\mathbf{x}, \mathbf{y} \in \mathbb{R}^{n-1}$ and $a, b \in \mathbb{R}$ these functions satisfy our assumption in,

$$\begin{aligned} h_{(1-\lambda)a+\lambda b}((1-\lambda)\mathbf{x} + \lambda\mathbf{y}) &= h((1-\lambda)\mathbf{x} + \lambda\mathbf{y}, (1-\lambda)a + \lambda b) \\ &= h((1-\lambda)(\mathbf{x}, a) + \lambda(\mathbf{y}, b)) \\ &\geq f(\mathbf{x}, a)^{1-\lambda} g(\mathbf{y}, b)^\lambda \\ &= f_a(\mathbf{x})^{1-\lambda} g_b(\mathbf{y})^\lambda \end{aligned}$$

So we are able to use our inductive hypothesis,

$$\int_{\mathbb{R}^{n-1}} h_{(1-\lambda)a+\lambda b} \geq \left(\int_{\mathbb{R}^{n-1}} f_a \right)^{1-\lambda} \left(\int_{\mathbb{R}^{n-1}} g_b \right)^\lambda$$

If we define the quantities,

$$F(s) = \int_{\mathbb{R}^{n-1}} f_s, \quad G(s) = \int_{\mathbb{R}^{n-1}} g_s, \quad H(s) = \int_{\mathbb{R}^{n-1}} h_s$$

Then the inequality above gives $H((1-\lambda)a + \lambda b) \geq (1-\lambda)F(a) + \lambda G(b)$ our antecedent on $\mathbb{R}$. Hence using the inductive hypothesis,

$$\int_{\mathbb{R}^n} h = \int_{\mathbb{R}} h \geq \left(\int_{\mathbb{R}} f \right)^{1-\lambda} \left(\int_{\mathbb{R}} g \right)^\lambda = \left(\int_{\mathbb{R}^n} f \right)^{1-\lambda} \left(\int_{\mathbb{R}^n} g \right)^\lambda$$

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Let $f_i \in L^{p_i}(\mathbb{R}^n)$ nonnegative functions where $p_i \geq 1$ and $\sum_i p_i^{-1} = 1$. We have the classic Holder's Inequality,

$$\int_{\mathbb{R}^n} \prod_{i=1}^m f_i \leq \prod_{i=1}^m \|f_i\|_{p_i} = \prod_{i=1}^m \left(\int_{\mathbb{R}^n} f_i^{p_i} \right)^{1/p_i}$$

For $\lambda \in (0, 1)$, in the particular case $m = 2$, $1/p_1 = 1 - \lambda$, $1/p_2 = \lambda$ applied to $f^{1-\lambda}, g^\lambda$ we have,

$$\int_{\mathbb{R}^n} f^{1-\lambda} g^\lambda \leq \left(\int_{\mathbb{R}^n} f \right)^{1-\lambda} \left(\int_{\mathbb{R}^n} g \right)^\lambda$$

By using the supremum of the l.h.s. the Prekopa-Leindler Inequality may also be written,

$$\int_{\mathbb{R}^n} \sup \{ f(\mathbf{x})^{1-\lambda} g(\mathbf{y})^\lambda \mid (1-\lambda)\mathbf{x} + \lambda\mathbf{y} = \mathbf{z} \} d\mathbf{z} \geq \left(\int_{\mathbb{R}^n} f \right)^{1-\lambda} \left(\int_{\mathbb{R}^n} g \right)^\lambda$$

Or, inductively,

$$\int_{\mathbb{R}^n} \sup \left\{ \prod_{i=1}^m f_i \mid \sum_{i=1}^m \frac{x_i}{p_i} = z_i \right\} d\mathbf{z} \geq \prod_{i=1}^m \|f_i\|_{p_i} = \prod_{i=1}^m \left(\int_{\mathbb{R}^n} f_i^{p_i} \right)^{1/p_i}$$

So the Prekopa-Leindler Inequality is in some sense the converse to Holder's Inequality.

We are now in a position to prove the Brunn-Minkowski Inequality in $\mathbb{R}^n$.

THM 4. BRUNN-MINKOWSKI INEQUALITY, EXPONENTIAL FORM

Let $\lambda \in (0, 1)$ and $X, Y \subseteq \mathbb{R}^n$ be bounded measurable sets s.t. $(1 - \lambda)X + \lambda Y$ is also measurable. Then,

$$\text{vol}((1 - \lambda)X + \lambda Y) \geq \text{vol}(X)^{1-\lambda} \text{vol}(Y)^\lambda$$

Proof.

Consider the functions $1_{(1-\lambda)X+\lambda Y}$, 1_X , and 1_Y and note that if $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ is s.t. $\mathbf{x} \in X$ and $\mathbf{y} \in Y$ then $(1 - \lambda)\mathbf{x} + \lambda\mathbf{y} \in (1 - \lambda)X + \lambda Y$. In particular we have the equality,

$$1_{(1-\lambda)X+\lambda Y}((1 - \lambda)\mathbf{x} + \lambda\mathbf{y}) = 1 \geq 1 = 1_X(\mathbf{x})^{1-\lambda} 1_Y(\mathbf{y})^\lambda$$

Therefore by Prekopa-Leindler's Inequality,

$$\begin{aligned} \text{vol}((1 - \lambda)X + \lambda Y) &= \int_{\mathbb{R}^n} 1_{(1-\lambda)X+\lambda Y} \\ &\geq \left(\int_{\mathbb{R}^n} 1_X \right)^{1-\lambda} \left(\int_{\mathbb{R}^n} 1_Y \right)^\lambda \\ &= \text{vol}(X)^{1-\lambda} \text{vol}(Y)^\lambda \end{aligned}$$

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And,

THM 5. BRUNN-MINKOWSKI INEQUALITY, STANDARD FORM

Let $\lambda \in (0, 1)$ and $X, Y \subseteq \mathbb{R}^n$ be bounded measurable sets s.t. $(1 - \lambda)X + \lambda Y$ is also measurable. Then,

$$\text{vol}((1 - \lambda)X + \lambda Y)^{1/n} \geq (1 - \lambda) \text{vol}(X)^{1/n} + \lambda \text{vol}(Y)^{1/n}$$

Proof.

From the previous inequality it suffices to choose the correct exponents λ' and sets X', Y' . Consider,

$$\lambda' = \frac{\text{vol}(\lambda Y)^{1/n}}{\text{vol}((1 - \lambda)X)^{1/n} + \text{vol}(\lambda Y)^{1/n}}, \quad 1 - \lambda' = \frac{\text{vol}((1 - \lambda)X)^{1/n}}{\text{vol}((1 - \lambda)X)^{1/n} + \text{vol}(\lambda Y)^{1/n}}$$

Hence by the previous inequality applied to arguments $X' = (1 - \lambda)X / \text{vol}((1 - \lambda)X)^{1/n}$, $Y' = \lambda Y / \text{vol}(\lambda Y)^{1/n}$,

$$\text{vol} \left(\frac{(1 - \lambda)X + \lambda Y}{\text{vol}((1 - \lambda)X)^{1/n} + \text{vol}(\lambda Y)^{1/n}} \right) \geq 1$$

So by positive-homogeneity (of degree n) of the Lebesgue measure,

$$\text{vol}((1 - \lambda)X + \lambda Y)^{1/n} \geq (1 - \lambda) \text{vol}(X)^{1/n} + \lambda \text{vol}(Y)^{1/n}$$

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Note first that outside of the interpretation of interpolation that the choices of λ and $1 - \lambda$ in the exponent are actually superfluous. In fact by homogeneity of the Lebesgue measure we generally have,

$$\text{vol}(sX + tY)^{1/n} \geq s \text{vol}(X)^{1/n} + t \text{vol}(Y)^{1/n}$$

There are in fact many ways to interpret the Brunn-Minkowski Inequality. One such way is with power means.

DEF 6. Note that the two forms of THM 4. and THM. 5 are in fact equivalent by Jensen's inequality for means. Namely for $-\infty \leq p \leq \infty$ define the 'power mean',

$$M_p(x_1, \dots, x_n) = \left(\frac{1}{n} \sum_{i=1}^n x_i^p \right)^{1/p}$$

Famous means of the argument $x_1, \dots, x_n$ correspond to choices for p . For instance $p = -\infty$ is the minimum, $p = -1$ is the harmonic mean, $p = 0$ is the geometric mean, $p = 1$ is the arithmetic mean, $p = 2$ the root mean square, and $p = +\infty$ the maximum.

The famous AM-GM inequality and all its variations are subsumed by the generalized mean inequality. Let $-\infty \leq p \leq q \leq +\infty$. Then,

$$M_p(x_1, \dots, x_m) \leq M_q(x_1, \dots, x_m)$$

With equality *iff* $x_1 = \dots = x_m$. THM 5. implying THM 4. is thus the choices of $p = 0$ and $q = 1/n$ with $x_1 = \text{vol}((1 - \lambda)X)$ and $x_2 = \text{vol}(\lambda Y)$.